Digital Audio-Video Signal Processing

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1 Question - 10 pts

Consider a discrete memoryless source X that emits symbols according to the binary alphabet $\{a, b\}$, with $p(a) = \Pr\{X = a\} = 0.1$.

- Compute the entropy of the source.
- Design a Huffman code for the source X (block size $N = 3$). Compute the average code length and compare it with the theoretical upper and lower bounds.

Consider the one realization of the source: *bbbbab*.

- Compute the number of bits/symbols spent when applying Huffman coding
- Compute the number of bits/symbols spent when applying Elias (arithmetic) coding. (You can assume negligible the cost of indicating the sequence length).

2 Question - 5 pts

Sketch the block diagram of a analysis-by-synthesis speech coder, illustrating the functionality of the individual blocks.

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X and Y denote two discrete memoryless sources and $L(x)$ the code length needed for representing the realization x.
   - [T] [F] $H(X|Y) \leq H(X)$
   - [T] [F] $H(X|Y) = H(X) \Leftrightarrow X$ and Y are statistically independent
   - [T] [F] $L(x) \geq H(X)$, $\forall x$
   - [T] [F] $H(X,Y) \geq H(X) + H(Y)$
2. Consider an image coding system
   - [T] [F] Entropy coding is lossy
   - [T] [F] Zonal quantization outperforms threshold quantization for non-stationary images
   - [T] [F] Luma and chroma components are typically coded with the same fidelity
   - [T] [F] In JPEG, the quantization matrix accounts for the sensitivity of the human visual system to the noise at different frequencies
3. Consider a perceptual audio coder
   - [T] [F] The analysis filter bank is the same used by the perceptual model
   - [T] [F] Noise and tonal maskers are characterized by different SMR
   - [T] [F] The critical bandwidth decreases at higher frequencies

4 Question - 12 pts

Load the signal x by means of the Matlab command `load bike_track`. The signal $x(n)$ represents the elevation track (in meters) collected during a bike ride, at a sampling time of $T = 5s$. You can consider x as a realization of a discrete source X with alphabet $A = \{0, 1, \ldots, 3000\}$

- Compute the empirical entropy of the source
- Use a simple uniform scalar quantizer followed by entropy coding to compress the source. Plot the rate-distortion function. (Hint: you are not asked to perform entropy coding. You can approximate the rate by means of the empirical entropy of the quantization levels)
- Use any algorithm of your choice to account for the source memory. Compute at least one rate-distortion point and compare it with the curve obtained before.